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A thermodynamic reassessment of the Al–Y system

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Abstract

The Al–Y system is reassessed based on a critical literature review involving recently measured phase diagram and thermodynamic data. In the thermodynamic optimization, the liquid phase is described by using a substitutional solution model. Al_3Y, AlY, Al_2Y_3 and AlY_2 are modeled as stoichiometric ones. A two-sublattice model $(Al, Y)_2(Al, Y)_1$ has been used to describe Al_2Y. A set of self-consistent thermodynamic parameters for the Al–Y binary system has been obtained by means of a CALPHAD approach applied to the selected experimental data. The reliable experimental phase diagram data and thermodynamic properties are well reproduced with the optimal thermodynamic parameters. The observed solidification paths for as-cast alloys can also be described reasonably using the present parameters. Significant improvement has been made, compared with previous assessments.
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1. Introduction

Numerous publications, both theoretical and experimental investigations, are available on synthesis, structure characterization and property measurements of amorphous phases during the past two decades. In particular, the amorphous alloys in the Al–Y–TM (TM = Fe, Co, Ni and Cu) systems have received extensive attention due to their unique mechanical properties, such as high tensile strength, good ductility, high corrosion resistance and thermal stability. Consequently, detailed investigation of their properties is of considerable interest not only with respect to their technological application but also from the point of view of thorough knowledge of their formation [1,2].

In order to improve the properties of the materials and design experiments most efficiently, knowledge of accurate phase diagram and thermodynamic data is of particular importance. Thermodynamic descriptions for lower order systems are an essential part of thermodynamic calculations for higher order systems. Therefore, obtaining the thermodynamic description about the Al–Y binary system is an important step in establishing a thermodynamic database for technologically important Al–Y–TM systems.

A critical literature review about the Al–Y binary system was carried out by Gschneidner and Calderwood in 1989 [3,4], whose phase diagram is generally accepted. Thermodynamic assessments for the system were conducted by two groups of authors [5,6]. However, a thermodynamic reassessment is still needed since (i) the calculated enthalpies of formation at 25 °C based on both thermodynamic assessments are too negative and obviously cannot match the experimental data well; (ii) too many parameters have been utilized to describe the liquid phase, which are 9 in the evaluation of Ran et al. [5] and 10 in Lukas's evaluation [6]; and (iii) plentiful reliable experimental phase diagram data have been published recently [7].

The purposes of the present work are to (i) clarify the discrepancy existing in the previous assessments of the Al–Y system and (ii) provide a self-consistent set of thermodynamic parameters for this binary system by carrying out a thermodynamic optimization of critically evaluated phase diagram and thermodynamic data.

2. Evaluation of experimental data

In this section, all experimental phase diagram and thermodynamic data available for the Al–Y binary system are evaluated. Because a detailed review on the phase diagram data has been given by Liu et al. [7] recently, the present authors just present a brief review on the phase diagram data and focus on

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Table 1
Summary of the phase diagram and thermodynamic data in the Al–Y system

Type of data	Reference	Experimental method	Remark[a]
Phase diagram:			
Whole phase diagram	[7]	XRD, DTA, metallography	+
Liquidus	[8]	Thermal analysis, XRD, metallography	+
Invariant reaction	[8]	Thermal analysis, XRD, metallography	Δ
Whole phase diagram	[9]	XRD, metallography, incipient-melting	Δ
Al-rich liquidus	[10,11]	emf	+
Al-rich eutectic point	[12]	Thermal analysis, optical microscopy	Δ
Al-rich liquidus	[13]	DTA, XRD, metallography	+
Al-rich eutectic point	[13]	DTA, XRD, metallography	Δ
ΔH_f at 25 °C:			
αAl_3Y, Al_2Y	[8]	Combustion calorimetry	+
AlY	[8]	Combustion calorimetry	▲
Al_2Y	[16]	High temperature calorimeter	+
Al_2Y_3	[17]	Direct synthesis calorimetry	+
αAl_3Y, Al_2Y, Al_2Y_3	[18]	Liquid-metal solution calorimetry	+
ΔH_{mix} at 1600 °C:			
Liquid	[14,15]	High temperature calorimeter	+
ΔH_f at 527 °C:			
αAl_3Y, Al_2Y, AlY, Al_2Y_3, AlY_2	[20]	emf	▲
$\Delta \overline{H}_Y$ at 940 °C:			
Liquid	[19]	High temperature solution calorimeter	Δ
ΔG_f at 527 °C:			
αAl_3Y, Al_2Y, AlY, Al_2Y_3, AlY_2	[20]	emf	▲

[a] Indicates whether the data are used or not used in the parameter optimization: +, used; Δ, not used but considered as reliable data for checking the modeling; ▲, not used. ΔH_f = enthalpy of formation; ΔH_{mix} = enthalpy of mixing; ΔG_f = Gibbs energy of formation; $\Delta \overline{H}_Y$ = partial enthalpy of formation for Y.

the evaluation of thermodynamic data. The sources for all data retained in the finally selected data set are listed in Table 1.

2.1. Phase diagram data

Most recently, the Al–Y binary phase diagram has been experimentally investigated by Liu et al. [7] using X-ray diffraction (XRD), differential thermal analysis (DTA), optical microscopy, and scanning electron microscopy with energy dispersive X-ray (SEM/EDX) techniques. Five intermetallic compounds Al_3Y, Al_2Y, AlY, Al_2Y_3, and AlY_2 have been synthesized. No Al_3Y_5 phase has been found. All the data published by Liu et al. [7] have been used in the present optimization because of a combination of several experimental methods.

The major contributions to the measurement of the Al–Y phase diagram before the work due to Liu et al. [7] are attributed to two groups of authors [8,9]. One is Snyder [8] who used thermal analysis, XRD and metallography to construct the phase diagram. The other is Lundin and Klodt [9], who employed XRD, metallography and incipient-melting observation to investigate the phase equilibria of this system. As shown in Table 2, the melting behaviors for the compounds and the invariant reaction types are in agreement with each other [8, 9]. However, the published data on several invariant reaction temperatures indicate noticeable discrepancies.

The solubility of Y in the Al melts from 665 to 837 °C was derived by Yamishchikov et al. [10,11] using electromotive force (emf) measurements. Utilizing thermal analysis and optical microscopy, Drits et al. [12] showed that the Al-rich eutectic point is located at 638 ± 3.4 °C and 3.08 at.% Y. By the combined use of DTA, metallography, and XRD techniques, Kononenko and Golubev [13] reported liquidus in the most Al-rich side and found the Al-rich eutectic point to be at 637 °C and 3.1 at.% Y.

The liquidus data from Snyder [8], the solubility data from Yamishchikov et al. [10,11] as well as the liquidus data from Kononenko and Golubev [13] have been used in the present optimization because these data are in reasonable agreement with each other. Other data [8,13] are only used to check the finally calculated phase diagram.

2.2. Thermodynamic data

The enthalpies of mixing for the liquid phase below 60 at.% Y at 1600 °C have been determined by Esin et al. [14] using a high temperature calorimeter. Following the same technique, Ryss et al. [15] measured the enthalpies of mixing of the melt at 1600 °C in the whole composition range. These enthalpy of mixing data [14,15] are included in the present modeling.

Several groups of investigators have measured the enthalpies of formation for intermetallics at 25 °C. By combustion calorimetry method, Snyder [8] reported the enthalpies of formation for the compounds Al_3Y, Al_2Y and AlY to be −47.1, −80.9 and −89.9 kJ/mol-atoms, respectively. Using a high temperature calorimeter, Jung et al. [16] determined the enthalpy of formation of Al_2Y to be −50.4±1.3 kJ/mol-atoms. Subsequently using direct synthesis calorimetry, Meschel and Kleppa [17] reported the enthalpy of formation for the compound Al_2Y_3 to be −40 kJ/mol-atoms. Using a

Table 2
Comparison between the measured [7–9,12,13] and calculated (this work) invariant reaction temperatures and the liquid compositions in the Al–Y system

Reaction	Liquid composition (at.% Y)	Temperature (°C)	Source
$L \Leftrightarrow (Al) + \alpha Al_3Y$		637 ± 2	[7]
	2.9	640	[8]
	3.3	650	[9]
	3	638 ± 3.4	[12]
	3	637	[13]
	2.96	637	This work
$\beta Al_3Y \Leftrightarrow \alpha Al_3Y, L, Al_2Y$		644	[8]
		644	This work
$L + Al_2Y \Leftrightarrow \beta Al_3Y$		980 ± 2	[7]
	11.7	980	[8]
	23	1355	[9]
	12.62	980	This work
$L \Leftrightarrow Al_2Y$	33.3	1490 ± 2	[7]
	33.3	1485	[8]
	33.3	1455	[9]
	33.3	1490	This work
$L + Al_2Y \Leftrightarrow AlY$		1138 ± 2	[7]
	52	1130	[8]
		1165	[9]
	51.1	1137	This work
$L \Leftrightarrow AlY + Al_2Y_3$		1081 ± 2	[7]
	58	1088	[8]
	57	1085	[9]
	56.2	1083	This work
$L \Leftrightarrow Al_2Y_3$	60	1105 ± 2	[7]
	60	1100	[8]
	60	1094	[9]
	60	1104	This work
$L + Al_2Y_3 \Leftrightarrow AlY_2$		977 ± 2	[7]
	75	985	[8]
	70	1035	[9]
	70.76	978	This work
$L \Leftrightarrow AlY_2 + (\alpha Y)$	≈75	955 ± 2	[7]
	75.3	960	[8]
		970	[9]
	73.7	954	This work
$(\beta Y) \Leftrightarrow (\alpha Y), L$	97.92	1478.84	This work

method of liquid-metal solution calorimetry, recently Timofeev et al. [18] found that the enthalpies of formation for Al_3Y, Al_2Y and Al_2Y_3 are −46.4 ± 1.8, −53.5 ± 2.5 and −46.9 ± 3.9 kJ/mol-atoms, respectively. All of these experimental values except the value for AlY published by Snyder [8] have been used in the optimization. The reported enthalpy of formation for AlY by Snyder [8] is too negative, compared with the other measurements.

Lee and Sommer [19] have investigated the partial enthalpies of mixing of Y in Al-rich (0.04–1.19 at.% Y) melts at 940 °C using a high temperature solution calorimeter. Kober et al. [20] have measured the enthalpies of formation and the Gibbs energies for the compounds Al_3Y, Al_2Y, AlY, Al_2Y_3 and AlY_2 at 527 °C by means of emf measurements. The experimental partial enthalpies of mixing data [19] will be compared with the calculated result from the finally obtained thermodynamic parameter. In a preliminary optimization, the enthalpies of formation and the Gibbs energies for the compounds [20] are included. It was realized that these data are not consistent with the other reliable thermodynamic and phase diagram data. As a consequence, these experimental data [20] are excluded from the final optimization.

3. Thermodynamic modeling

The Gibbs energy function ${}^oG_i^{\varphi}(T) = G_i^{\varphi}(T) - H_i^{SER}$ for element i (i = Al, Y) in the phase φ (φ = liquid, (Al), (αY), (βY)) is described by an equation of the form:

$$ {}^oG_i^{\varphi}(T) = a + b \cdot T + c \cdot T \cdot \ln(T) + d \cdot T^2 + e \cdot T^{-1} + f \cdot T^3 + g \cdot T^7 + h \cdot T^{-9} \quad (1) $$

where H_i^{SER} is the molar enthalpy of the element i at 25 °C and 1 bar in its standard element reference (SER) state, and T is the absolute temperature. In the present modeling, the Gibbs energies for the pure elements are taken from the compilation by Dinsdale [21].

3.1. The liquid, (Al), (αY), (βY) phases

The liquid, (Al), (αY), and (βY) solution phases are described by Redlich–Kister polynomials [22], and the Gibbs energy for the liquid is expressed by the following equation:

$$G_m^L - H^{SER} = {}^{\mathrm{ref}}G_m^L + {}^{\mathrm{ideal}}G_m^L + {}^{ex}G_m^L$$

with ${}^{\mathrm{ref}}G_m^L = (1-x)\,{}^oG_{\mathrm{Al}}^L + x\,{}^oG_{\mathrm{Y}}^L$, ${}^{\mathrm{ideal}}G_m^L = RT[x\ln x + (1-x)\ln(1-x)]$, and

$$ {}^{ex}G_m^L = x(1-x)[a_0 + b_0T + (1-2x)(a_1 + b_1T) + \ldots] \quad (2)$$

in which $H^{SER}$is the abbreviation of $(1-x)H_{\mathrm{Al}}^{SER} + xH_{\mathrm{Y}}^{SER}$, R the gas constant, and x the mole fraction of Y. The interaction parameters a_0, b_0, a_1 and b_1 are to be optimized from the experimental phase diagram and thermodynamic data.

The (Al), (αY), and (βY) are modeled as ideal solution. No excess Gibbs energy terms are needed since the mutual solubilities between Al and Y are reported to be negligible.

3.2. The compounds AlY, Al_2Y_3 and AlY_2

All these phases are modeled as stoichiometric phases, and the Gibbs energy of each compound $Al_{1-x}Y_x$ is given by the following expression:

$$G_{\mathrm{Al:Y}}^{\mathrm{Al}_{1-x}\mathrm{Y}_x} - H^{SER} = {}^{\mathrm{ref}}G_m^{\varphi} + {}^{f}G_m^{\varphi}$$

with ${}^{\mathrm{ref}}G_m^{\varphi} = (1-x)\,{}^oG_{\mathrm{Al}}^{\mathrm{fcc}} + x\,{}^oG_{\mathrm{Y}}^{\mathrm{hcp}}$

and $\quad {}^{f}G_m^{\varphi} = A_x + B_xT. \quad (3)$

The coefficients A_x and B_x can be evaluated by using thermodynamic and phase diagram data.

3.3. The compound Al_3Y (αAl_3Y and βAl_3Y)

For Al_3Y, as discussed by Liu et al. [7], there are two polymorphic forms, αAl_3Y (low-temperature form) and βAl_3Y (high-temperature form), and the phase transformation temperature between αAl_3Y and βAl_3Y is reported to be 644 °C [8]. Both αAl_3Y and βAl_3Y are modeled as stoichiometric phases. For αAl_3Y, the Gibbs energy is described by an analogous equation as Eq. (3). For βAl_3Y, the Gibbs energy is expressed as follows,

$$G_{\mathrm{Al:Y}}^{\beta\mathrm{Al}_{0.75}\mathrm{Y}_{0.25}} - G_{\mathrm{Al:Y}}^{\alpha\mathrm{Al}_{0.75}\mathrm{Y}_{0.25}} = -917.15C_x + C_xT \quad (4)$$

in which the coefficient C_x can be optimized from the experimental phase diagram data. At the transition temperature 917.15 K between αAl_3Y and βAl_3Y, the Gibbs energies of these two phases are equal.

3.4. The compound Al_2Y

The crystal structure of Al_2Y is of C15 laves type, which is currently described with a two-sublattice model $(\mathrm{Al,Y})_2(\mathrm{Al,Y})_1$. According to the general sublattice model [23, 24], the Gibbs energy of this phase per mol-atoms is given by an equation of the form:

$$\begin{aligned} G_m^{\mathrm{Al}_2\mathrm{Y}} - H^{SER} = {} & y'_{\mathrm{Al}} \cdot y''_{\mathrm{Al}} \cdot G_{\mathrm{Al:Al}}^{0,\mathrm{Al}_2\mathrm{Y}} + y'_{\mathrm{Al}} \cdot y''_{\mathrm{Y}} \cdot G_{\mathrm{Al:Y}}^{0,\mathrm{Al}_2\mathrm{Y}} \\ & + y'_{\mathrm{Y}} \cdot y''_{\mathrm{Al}} \cdot G_{\mathrm{Y:Al}}^{0,\mathrm{Al}_2\mathrm{Y}} + y'_{\mathrm{Y}} \cdot y''_{\mathrm{Y}} \cdot G_{\mathrm{Y:Y}}^{0,\mathrm{Al}_2\mathrm{Y}} \\ & + \frac{2}{3} \cdot R \cdot T(y'_{\mathrm{Al}} \cdot \ln y'_{\mathrm{Al}} + y'_{\mathrm{Y}} \ln y'_{\mathrm{Y}}) \\ & + \frac{1}{3} \cdot R \cdot T(y''_{\mathrm{Al}} \cdot \ln y''_{\mathrm{Al}} + y''_{\mathrm{Y}} \ln y''_{\mathrm{Y}}) \\ & + y'_{\mathrm{Al}} \cdot y'_{\mathrm{Y}} \cdot y''_{\mathrm{Al}} \cdot L_{\mathrm{Al,Y:Al}}^{0,\mathrm{Al}_2\mathrm{Y}} \\ & + y'_{\mathrm{Al}} \cdot y'_{\mathrm{Y}} \cdot y''_{\mathrm{Y}} \cdot L_{\mathrm{Al,Y:Y}}^{0,\mathrm{Al}_2\mathrm{Y}} \\ & + y'_{\mathrm{Al}} \cdot y''_{\mathrm{Al}} \cdot y''_{\mathrm{Y}} \cdot L_{\mathrm{Al:Al,Y}}^{0,\mathrm{Al}_2\mathrm{Y}} \\ & + y'_{\mathrm{Y}} \cdot y''_{\mathrm{Al}} \cdot y''_{\mathrm{Y}} \cdot L_{\mathrm{Y:Al,Y}}^{0,\mathrm{Al}_2\mathrm{Y}} + \ldots \end{aligned} \quad (5)$$

in which y'_{Al} and y'_{Y} are the site fractions of Al and Y on the first sublattice, and y''_{Al} and y''_{Y} on the second one. The four parameters denoted $G_{*:*}^{0,\mathrm{Al}_2\mathrm{Y}}$ (also called compound energies, $*$ denotes Al or Y) are expressed relative to the Gibbs energies of pure (Al) and (αY) at the same temperature. The L^0 parameters represent the interactions primarily within each sublattice. Since the experimental data involving Al_2Y are limited, it would be difficult to assess the quantities in Eq. (5) directly. In the present work, Al_2Y is first modeled as stoichiometric compounds, then with a two-sublattice model $(\mathrm{Al,Y})_2(\mathrm{Al,Y})_1$.

4. Result and discussion

Evaluation of the model parameters is attained by recurrent runs of the PARROT program [25], which works by minimizing the square sum of the differences between measured and calculated values. The step-by-step optimization procedure carefully described by Du et al. [26] was utilized in the present assessment.

The optimization began with the liquid phase. For liquid, at least a_0 and b_0 in Eq. (2) can be adjusted because the liquidus and the enthalpy of mixing have been measured accurately over the whole composition range. Since the experimental enthalpy of mixing versus mole fraction is asymmetric, as shown in Fig. 1, at least two coefficients (a_0 and a_1) are needed to describe the asymmetric feature. As for the entropy, it is impossible to tell in advance from the experimental phase diagram data whether more than one coefficient is necessary. Consequently, the optimization was started with one coefficient (b_0). In the present case, it was found that a_2, a_3, b_1 and b_2 should also be introduced in order to describe the properties of the liquid phase satisfactorily.

Secondly, the intermetallic phases were included in the optimization one by one. For αAl_3Y, the enthalpy of formation has been measured by two groups of investigators [8,18]. This quality is directly expressed by Eq. (3). The coefficient A_0 was set equal to the mean value of Snyder [8] and Timofeev et al. [18] and was adjusted in the thermodynamic modeling. For Al_2Y, the enthalpy of formation has been measured by Snyder [8], Jung et al. [16] and Timofeev et al. [18]. In the preliminary optimization, the mean value

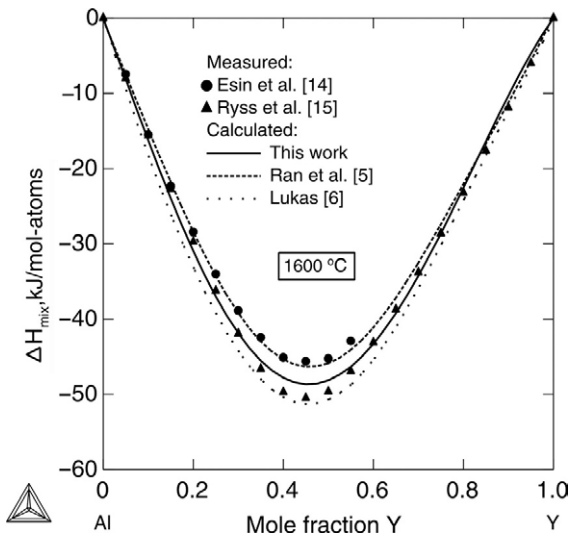

Fig. 1. Calculated enthalpies of mixing for the liquid phase at 1600 °C together with the experimental data [14,15].

of the investigation [16,18] was used as input. It was realized the assessed mean value was not consistent with other reliable thermodynamic and phase diagram data. Therefore, the mean value of the three measurements [8,16] and [18] was used as input for the final thermodynamic optimization. To reduce the number of adjustable coefficients, the compound energies in Eq. (5) are subjected to the following relationship [27,28]:

$$G^{0,\mathrm{Al_2Y}}_{\mathrm{Y:Al}} = -G^{0,\mathrm{Al_2Y}}_{\mathrm{Al:Y}} + G^{0,\mathrm{Al_2Y}}_{\mathrm{Al:Al}} + G^{0,\mathrm{Al_2Y}}_{\mathrm{Y:Y}}. \quad (6)$$

The quantities $G^{0,\mathrm{Al_2Y}}_{\mathrm{Al:Al}}$ and $G^{0,\mathrm{Al_2Y}}_{\mathrm{Y:Y}}$ in Eqs. (5) and (6) describe a metastable form of pure Al and Y with C15 laves structure. These parameters are fixed to be 5000 J/mol-atoms relative to the stable (Al) and (αY) structure. Experimental work [7–9] indicated a negligible homogeneity range for Al_2Y. This feature is well accounted for by adopting just one interaction parameter, $L^{0,\mathrm{Al_2Y}}_{\mathrm{Al,Y:Y}} = L^{0,\mathrm{Al_2Y}}_{\mathrm{Al:Al,Y}} = L^{0,\mathrm{Al_2Y}}_{\mathrm{Y:Al,Y}} = L^{0,\mathrm{Al_2Y}}_{\mathrm{Al,Y:Al}} = 6667$. Therefore, there are only two parameters for Al_2Y to be optimized.

In the case of Al_2Y_3, the enthalpy of formation has been measured by Meschel and Kleppa [17] and Timofeev et al. [18]. The average value of them [17,18] was adopted in the thermodynamic optimization. For the other two compounds (AlY, AlY_2), there are no experimental data about their enthalpies of formation. The data for these two compounds were reasonably estimated from the line connecting those of Al_2Y and Al_2Y_3 as well as Al_2Y_3 and Y.

Finally, the thermodynamic parameters for all the phases were optimized simultaneously by taking into account all of the selected phase diagram and thermodynamic data. The thermodynamic parameters finally obtained in the present work include 18 coefficients, 7 of which are for liquid, and 11 of which are for the intermetallics (Al_3Y, Al_2Y, AlY, Al_2Y_3 and AlY_2). These optimized coefficients are listed in Table 3. It should be mentioned that the number of the parameters used for the liquid phase in the present work is less than that of previous optimization [5,6], in which it is 9 and 10, respectively. Any attempt to reduce the number of the thermodynamic parameters used for the liquid phase resulted in worse agreement with the experimental phase diagram data.

Table 3
Optimized thermodynamic parameters for the Al–Y system

Stoichiometric model:
${}^{f}G^{\varphi}_{m} = A_x + B_x T$ (${}^{f}G^{\varphi}_{m}$ is Gibbs energy of formation)

Phase	A_x, J/mol-atoms	B_x, J/(mol-atoms K)
AlY	−54 911.9	5.48155
Al_2Y_3	−48 406.3	4.82944
AlY_2	−44 886.6	7.39566
αAl_3Y	−49 411.4	5.29096

$G^{\beta \mathrm{Al}_{0.75}\mathrm{Y}_{0.25}}_{\mathrm{Al:Y}} - G^{\alpha \mathrm{Al}_{0.75}\mathrm{Y}_{0.25}}_{\mathrm{Al:Y}} = -917.15C_x + C_x T$

Phase	C_x, J/(mol-atoms K)
βAl_3Y	−0.19096

Sublattice model $(\mathrm{Al, Y})_2(\mathrm{Al, Y})_1$ for Al_2Y

Parameter	Value, J/mol-atoms
$G^{0,\mathrm{Al_2Y}}_{\mathrm{Al:Al}} - G^{(\mathrm{Al})}_{\mathrm{Al}}$	5000
$G^{0,\mathrm{Al_2Y}}_{\mathrm{Y:Y}} - G^{(\alpha\mathrm{Y})}_{\mathrm{Y}}$	5000
$G^{0,\mathrm{Al_2Y}}_{\mathrm{Al:Y}} - \left(\frac{2}{3}G^{(\mathrm{Al})}_{\mathrm{Al}} + \frac{1}{3}G^{(\alpha\mathrm{Y})}_{\mathrm{Y}}\right)$	$-66\,528.4 + 9.42976T$
$G^{0,\mathrm{Al_2Y}}_{\mathrm{Y:Al}} - \left(\frac{2}{3}G^{(\alpha\mathrm{Y})}_{\mathrm{Y}} + \frac{1}{3}G^{(\mathrm{Al})}_{\mathrm{Al}}\right)$	$76\,528.4 - 9.42976T$
$L^{0,\mathrm{Al_2Y}}_{\mathrm{Al,Y:Y}} = L^{0,\mathrm{Al_2Y}}_{\mathrm{Al:Al,Y}} = L^{0,\mathrm{Al_2Y}}_{\mathrm{Y:Al,Y}} = L^{0,\mathrm{Al_2Y}}_{\mathrm{Al,Y:Al}}$	6667

Redlich–Kister formalism:
${}^{ex}G^{L}_{m} = x(1-x)[a_0 + b_0 T + (1-2x)(a_1 + b_1 T) + \cdots]$
(${}^{ex}G^{L}_{m}$ is excess Gibbs energy)

Phase	i	a_i, J/mol-atoms	b_i, J/mol-atoms K
Liquid	0	−192 571.2	29.03622
	1	−46 742.9	−0.46159
	2	70 259.0	−28.99990
	3	9002.8	

Table 2 compares the calculated and measured invariant equilibria in the Al–Y system. The differences between the calculated and measured temperatures [7] are within 2 °C for all of the invariant reactions. In Fig. 1, the calculated enthalpies of mixing for the liquid phase at 1600 °C are compared with the experimental values [14,15]. The agreement using the present modeling is judged satisfactorily.

In Fig. 2, the calculated enthalpies of formation for the compounds at 25 °C and the experimental data are shown. The present parameters can describe recent experimental data [17, 18] very well. This is not the case for the previous ones [5,6].

In Fig. 3, the calculated phase diagram is compared with the experimental data points from the literature [7–13]. The presently computed phase diagram is in good agreement with the experimental data. The Pandat [29] program has been used to check if the calculated phase diagram is a real stable one. The calculated phase diagrams coincide with each other by using both Thermo-Calc and Pandat programs.

Fig. 4 compares the calculated partial enthalpies of mixing of Y in the Al-rich side at 940 °C with the experimental values of Lee and Sommer [19]. The scattering experimental data are

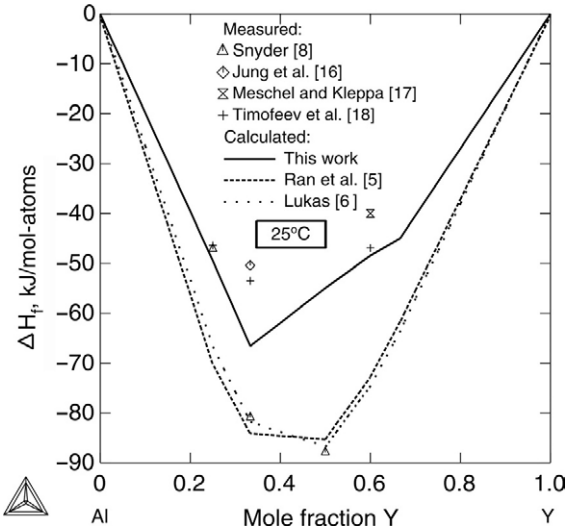

Fig. 2. Calculated enthalpies of formation for the compounds at 25 °C and the corresponding experimental data [8,16–18].

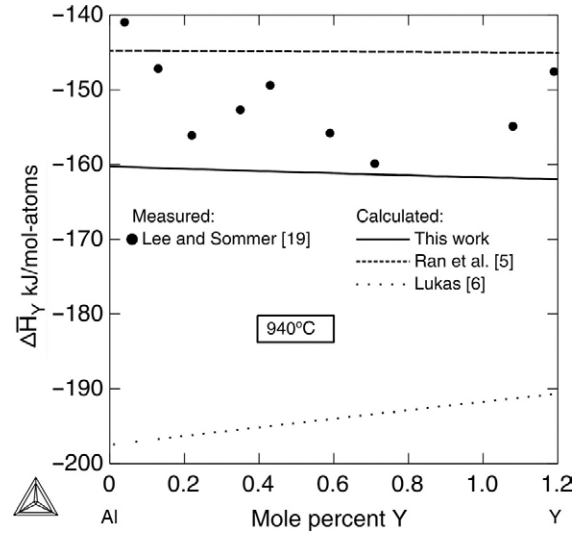

Fig. 4. Predicted partial enthalpies of mixing of Y in the Al-rich side at 940 °C with the experimental values from Lee and Sommer [19].

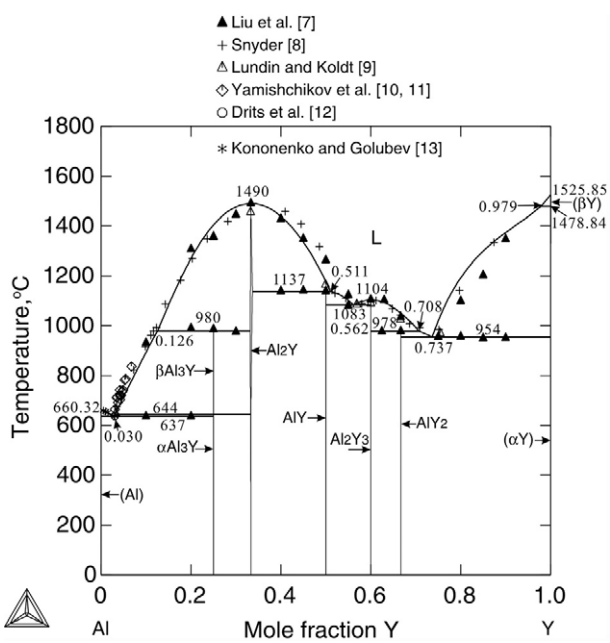

Fig. 3. Calculated Al–Y phase diagram with the experimental data [7–13].

reasonably described by the present thermodynamic parameters and those of Ran et al. [5].

Using the Gulliver–Scheil model [30,31] and thermodynamic parameters of Lukas [6], Liu et al. [7] calculated the solidification path for the as-cast alloy $Al_{0.37}Y_{0.63}$ (atomic fraction). The computed solidification path confirms the observed as-cast microstructure. However, there are some differences between the calculated phase transition temperatures and the experimental ones. Based on the present thermodynamic parameters, the solidification path for the same alloy has been calculated. As shown in Fig. 5, the present modeling can predict the as-cast microstructure and their relative amounts reasonably. In addition, the calculated phase transition temperatures agree well with the observed ones [7].

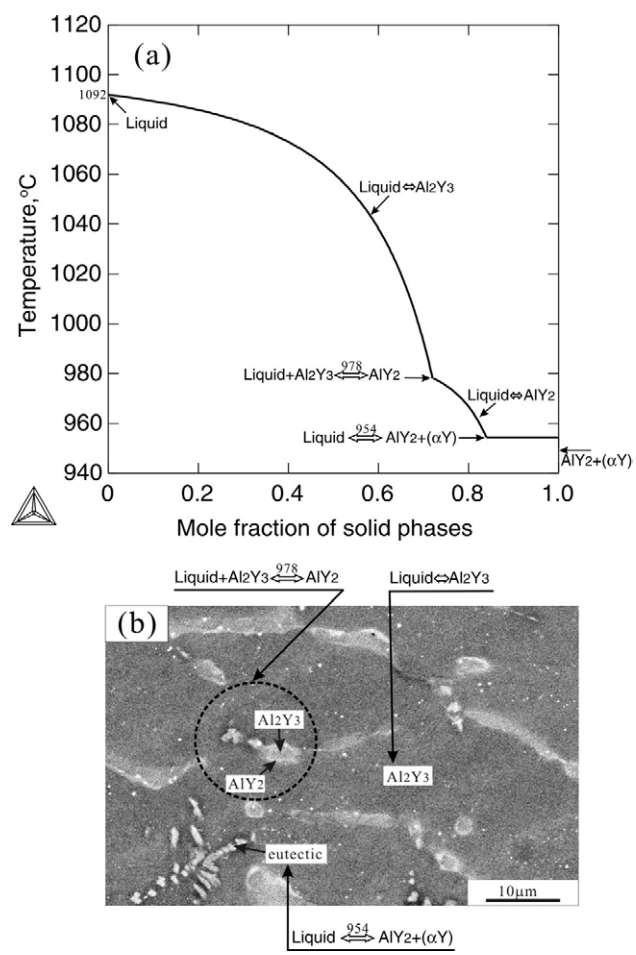

Fig. 5. Calculated solidification path for the as-cast alloy $Al_{0.37}Y_{0.63}$ (atomic fraction) and the observed microstructure from Liu et al. [7]. (a) Calculated solidification path; (b) observed microstructure [7].

The currently obtained Al–Y phase diagram is shown in Fig. 6 for easy perception without the experimental data points

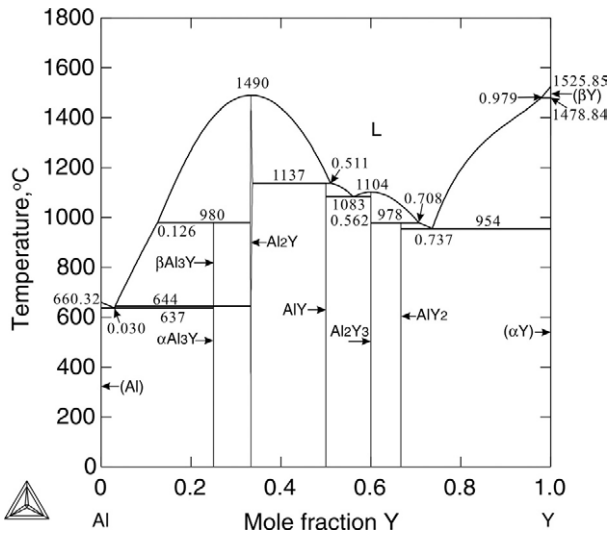

Fig. 6. Revised Al–Y phase diagram in accordance with the present work.

and in the practical Celsius range. Due to the negligible solubilities for Al in both (αY) and (βY), the invariant equilibrium among liquid, (αY) and (βY) is of degenerated feature. This diagram is expected to substitute for the currently accepted version [4,6].

The thermodynamic parameters for the Al–Y system obtained in the present work have been incorporated into a thermodynamic database of the multicomponent Al-based system [32–34].

5. Summary

An optimal thermodynamic data set for the Al–Y system has been obtained by considering the newly published experimental phase diagram data as well as critically evaluated literature data. The comprehensive comparisons show that the calculated phase diagram and thermodynamic properties are in good agreement with the experimental information. Compared with the previous assessments, the present modeling utilizes less parameters for the liquid phase and the calculated enthalpies of formation are more reasonable. The obtained thermodynamic parameters can also be used to simulate the solidification paths for the as-cast alloys.

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